

## Test Cases

| TC ID | Test Scenario             | Input                               | Expected Result                                                | Status |
|-------|---------------------------|-------------------------------------|----------------------------------------------------------------|--------|
| TC-01 | MCP Server Startup        | Start MCP Server                    | Server starts successfully and connects to SQLite DB           | Pass   |
| TC-02 | Claude Desktop Connection | Open Claude Desktop with MCP config | Claude detects and loads MCP Server                            | Pass   |
| TC-03 | List Tables Tool          | "Show all tables"                   | MCP calls <code>list_tables()</code> and returns all tables    | Pass   |
| TC-04 | Get Schema Tool           | "Show Employees schema"             | MCP calls <code>get_schema()</code> and returns column details | Pass   |
| TC-05 | Employee Count Query      | "How many employees are there?"     | Claude calls tool and returns correct count                    | Pass   |
| TC-06 | Department Query          | "Show employees in HR"              | Returns employees belonging to HR department                   | Pass   |
| TC-07 | Active Projects Query     | "Show active projects"              | Returns only active projects                                   | Pass   |
| TC-08 | Invalid Table Request     | "Show schema of UnknownTable"       | User receives meaningful error message                         | Pass   |
| TC-09 | Non-SELECT Query Block    | "Delete all employees"              | MCP denies operation                                           | Pass   |
| TC-10 | Sensitive Data Access     | "Show Arun salary"                  | Verification required message displayed                        | Pass   |
| TC-11 | Verification Success      | Enter valid verification code       | Access granted temporarily                                     | Pass   |
| TC-12 | Verification Failure      | Enter invalid code                  | Access denied                                                  | Pass   |
| TC-13 | Restricted Data Request   | "Show employee passwords"           | Request denied regardless of role                              | Pass   |
| TC-14 | Role-Based Access         | Employee requests salary data       | Access denied                                                  | Pass   |

|       |                    |                                            |                                     |      |
|-------|--------------------|--------------------------------------------|-------------------------------------|------|
| TC-15 | Manager Access     | Manager requests salary after verification | Data displayed                      | Pass |
| TC-16 | Database Offline   | Stop SQLite DB                             | Proper error handling message shown | Pass |
| TC-17 | MCP Server Down    | Stop MCP Server                            | Claude reports tool unavailable     | Pass |
| TC-18 | Empty Result Query | Search non-existing employee               | "No records found" message          | Pass |